Separating Mean Arterial Pressure- and Carbon Dioxide-Associated Contributions to Cerebral Blood Flow Velocity in Mild Cognitive Impairment Using Multivariable Transfer Function Analysis

[Author names and affiliations to be added]

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DRAFT NOTE: Confirm the final title, author order, affiliations, corresponding author, target journal, and whether the journal prefers "cerebral blood velocity" or "cerebral blood flow velocity." This draft uses "associated" instead of "mediated" because the observational transfer functions do not establish causality.

Abstract

Background: Dynamic cerebral autoregulation is commonly assessed with single-input, single-output (SISO) transfer function analysis using mean arterial pressure (MAP) as the input and cerebral blood flow velocity (CBFV) as the output. CBFV is also affected by carbon dioxide, so a MAP-only model may mix pressure-associated and carbon-dioxide-associated dynamics.

Objective: We tested whether simultaneously modeling MAP and end-tidal carbon dioxide ($P_{\mathrm{ET}}CO_2$) materially changed pathway-specific transfer function estimates and whether the resulting multivariable estimates differed between cognitively normal controls (NC) and participants with mild cognitive impairment (MCI).

Methods: Baseline MAP, $P_{\mathrm{ET}}CO_2$, and CBFV recordings from 23 NC and 52 MCI participants were resampled to 4 Hz. Auto- and cross-spectra were estimated using 128-second Hann windows with 50% overlap and at least three Welch windows. Separate SISO models and a two-input, single-output (MISO) model were calculated over 0–0.35 Hz. Prespecified bands were 0.005–0.024, 0.024–0.070, 0.070–0.200, and

0.200–0.350 Hz. Gain comparisons used paired tests within participants; NC–MCI comparisons used unequal-variance tests. Phase was analyzed with circular statistics and permutation tests. The Benjamini–Hochberg (BH) method controlled the false discovery rate within prespecified analysis families.

Results: Simultaneous modeling changed gain estimates in both groups. MISO MAP gain was lower than SISO MAP gain in the very-very-low-frequency band in NC (mean difference −0.240, 95% confidence interval [CI] −0.384 to −0.096, BH-adjusted $P = 0.0052$) and MCI (−0.183, 95% CI −0.290 to −0.076, BH-adjusted $P = 0.0038$). The larger and more consistent changes occurred in the carbon dioxide pathway. MISO CO_2 gain was lower than SISO CO_2 gain in the low- and high-frequency bands in NC and in the very-low-, low-, and high-frequency bands in MCI (all BH-adjusted $P < 0.006$). No prespecified band-level NC–MCI difference in MISO gain, partial coherence, phase, or MAP-minus-CO_2 pathway balance survived BH correction. An exploratory frequency-wise analysis identified higher NC than MCI MISO MAP gain at 0.1484 Hz (BH-adjusted $P = 0.0366$ within that frequency curve).

Conclusions: Modeling MAP and $P_{\mathrm{ET}}CO_2$ together materially changed pathway-specific gain estimates, particularly for the CO_2 pathway. These results support the methodological premise that MAP-only and CO_2-only SISO estimates are not interchangeable with conditional MISO estimates. However, the present data did not support the prespecified hypothesis of a broad band-level difference in conditional cerebral hemodynamics between NC and MCI. The isolated 0.1484-Hz group difference is hypothesis-generating and requires confirmation.

Keywords: cerebral autoregulation; carbon dioxide; mild cognitive impairment; transcranial Doppler; transfer function analysis; multivariable spectral analysis; coherence

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1 Introduction

The brain requires a continuous supply of oxygen and nutrients, but systemic blood pressure changes from moment to moment. Cerebral autoregulation is one part of the broader set of mechanisms that helps maintain cerebral perfusion despite these changes [1]. Dynamic cerebral autoregulation (dCA) focuses on the response of cerebral blood flow to relatively rapid or spontaneous changes in arterial pressure. In human studies, dCA is often estimated from continuous MAP and transcranial Doppler-derived CBFV signals. Transfer function analysis (TFA) provides a frequency-specific description of this pressure–flow relationship [2].

Cerebrovascular regulation is relevant to mild cognitive impairment (MCI) because vascular dysfunction and reduced perfusion may contribute to the development or progression of cognitive decline [3, 4]. At the same time, the existing dCA literature in MCI is mixed. Individual studies have often found limited or no group differences in conventional TFA gain and phase [5, 6], and a recent systematic review did not find consistent evidence of poorer spontaneous dCA in MCI or Alzheimer's disease [7]. These mixed results may reflect true biological heterogeneity, limited sample sizes, different experimental conditions, or limitations in how the input–output system is represented.

Standard TFA usually treats MAP as the only input to CBFV. This is useful and well established, but it is incomplete. Carbon dioxide is a strong and rapid regulator of cerebral vascular tone, and even small changes in arterial carbon dioxide can produce substantial changes in cerebral blood flow [8]. End-tidal carbon dioxide is not identical to arterial carbon dioxide, but it is commonly used as a practical continuous surrogate. If MAP and $P_{\mathrm{ET}}CO_2$ fluctuate in overlapping frequency ranges, a MAP-only SISO transfer function can contain CBFV variation associated with both MAP and CO_2. The same problem applies in reverse to a CO_2-only SISO model.

Prior multivariable work provides a basis for addressing this problem. Panerai and colleagues modeled spontaneous MAP and $P_{\mathrm{ET}}CO_2$ fluctuations together [9]. Peng and colleagues later showed that including gas-related inputs increased multiple coherence below 0.05 Hz and could alter the low-frequency MAP-to-CBFV transfer function [10]. Their result does not mean that every multivariable estimate is automatically more accurate. It shows that the univariable estimate and the conditional multivariable estimate answer different questions. The SISO model describes the total pairwise association between one input

and CBFV. The MISO model estimates the association of each input with CBFV while accounting for the other measured input.

This distinction is important for MCI. A MAP-only group comparison may miss or misstate a pressure-associated difference if the groups also differ in CO_2 dynamics or in MAP–CO_2 coupling. Conversely, a multivariable model may show that an apparent SISO difference is shared with the second input. The purpose of the present study was therefore not simply to maximize coherence. It was to determine whether simultaneous MAP and $P_{\mathrm{ET}}CO_2$ modeling materially changes pathway-specific interpretation, and then to use that model to compare NC and MCI participants.

1.1 Central hypothesis and aims

Our central hypothesis was that simultaneously modeling MAP and $P_{\mathrm{ET}}CO_2$ would materially alter pathway-specific transfer function estimates relative to separate SISO models, particularly where the two inputs share frequency content, and that these conditional estimates could reveal whether MAP- and CO_2-associated CBFV dynamics differ between NC and MCI.

The aims were distinct tests of this hypothesis:

1. **Model dependence and pathway separation.** We compared MISO and SISO gain and phase within the same participants. We expected the largest changes where one-input estimates contained information shared with the second input.

2. **Input coupling.** We tested whether MAP–CO_2 input coherence was associated with the absolute size of the MISO-minus-SISO gain change. This analysis was methodological and exploratory because the two quantities are derived from the same spectral matrix.

3. **Main biological comparison.** We compared NC and MCI using MISO MAP gain, MAP partial coherence, and circular phase after accounting for $P_{\mathrm{ET}}CO_2$. Parallel CO_2-pathway comparisons tested whether any group pattern was pressure-specific.

4. **Pathway balance.** We compared the within-subject difference between MAP and CO_2 partial coherence across groups. Positive values indicate a stronger conditional MAP–CBFV association than conditional CO_2–CBFV association in that band.

Multiple coherence was retained as a descriptive measure of how much CBFV power was linearly represented in the fitted two-input model. It was not treated as proof that MISO predicts unseen data better than SISO. Such a claim would require cross-validation on recordings long enough to estimate a model and evaluate it on an independent segment.

2 Methods

2.1 Study design and participants

This was a cross-sectional secondary analysis of baseline physiological recordings from NC and MCI participants. The final TFA cohort contained 23 NC and 52 MCI participants. All 75 recordings met the analysis requirement of at least three Welch windows after preprocessing. Demographic information was available for all NC participants and 51 of 52 MCI participants; MCI 302 was present in the physiological dataset but did not have a matching row in the available demographic workbook.

The available cognitive descriptors included the Mini-Mental State Examination (MMSE) [18] and Montreal Cognitive Assessment (MoCA) [19].

DRAFT NOTE: Add the parent study name, recruitment setting, exact NC and MCI diagnostic criteria, inclusion and exclusion criteria, institutional review board approval number, informed-consent language, and whether diagnosis was amnestic, non-amnestic, single-domain, or multidomain MCI. Confirm how CDR, MoCA, neuropsychological testing, and clinical adjudication were used. Do not infer these details from the spreadsheet alone.

2.2 Physiological recording and signal preparation

Continuous arterial pressure, middle cerebral artery CBFV, end-tidal carbon dioxide, and electrocardiography were acquired during resting baseline. The source data were recorded at 500 Hz in the acquisition system and converted before the present analysis into time-aligned beat-to-beat MAP and CBFV and breath-to-breath $P_{\mathrm{ET}}CO_2$ values. Cardiac R peaks defined the beat intervals. The available beat-level signals were visually reviewed before export. No short artifacts required interpolation in the recordings included here.

DRAFT NOTE: Add the recording posture, baseline duration target, arterial pressure device and calibration, transcranial Doppler manufacturer and probe frequency, insonated artery and side, probe fixation method, capnograph model, sampling hardware/software versions, and room/breathing conditions. Confirm whether CBFV was unilateral or averaged across hemispheres.

2.3 Preprocessing

The MATLAB workflow first removed rows containing non-finite values in any of the time, MAP, $P_{\mathrm{ET}}CO_2$, or CBFV signals. Some exported $P_{\mathrm{ET}}CO_2$ columns contained leading zeros and a large startup transition before stable values were present. The software removed this startup portion automatically so that columns did not need to be edited manually. It identified the first nonzero $P_{\mathrm{ET}}CO_2$ value, calculated absolute consecutive changes in the remaining signal, and defined unusually large changes using the mean plus one standard deviation of the upper 10% of changes. Samples were removed from the start until a nonzero value followed by a change below this data-derived threshold was reached.

After startup removal, all signals were linearly resampled to 4 Hz. Mean MAP, mean $P_{\mathrm{ET}}CO_2$, and mean CBFV were stored before centering. CBFV was then normalized to percent of its recording mean,

$$\mathrm{CBFV}_{\%}(t) = 100\frac{\mathrm{CBFV}(t)}{\overline{\mathrm{CBFV}}}. \tag{1}$$

The mean was removed from MAP, $P_{\mathrm{ET}}CO_2$, and normalized CBFV before spectral estimation. Mean removal eliminates the constant, zero-frequency component so that the spectra describe fluctuations around each participant's baseline. Polynomial detrending was available in the software but disabled for the present analysis to avoid imposing an additional low-frequency transformation.

2.4 Welch spectral estimation

Auto- and cross-spectra were estimated using Welch's method [11]. Each segment used a 128-second Hann window with 50% overlap. At 4 Hz, the window length was

$$L = 128\ \mathrm{s} \times 4\ \mathrm{samples/s} = 512\ \mathrm{samples}, \tag{2}$$

with an overlap of 256 samples and a step of 256 samples. Therefore, the minimum number of points required for three windows was

$$N_{\min} = L + (K-1)(L - L_{\text{overlap}}) = 512 + 2(256) = 1024 \text{ samples}, \tag{3}$$

which corresponds to 256 seconds at 4 Hz. The fast Fourier transform length was 512 samples, giving a frequency spacing of

$$\Delta f = \frac{f_s}{N_{\text{FFT}}} = \frac{4}{512} = 0.0078125 \text{ Hz}. \tag{4}$$

The current full-frequency outputs covered 0–0.35 Hz. With the available frequency spacing, the last included frequency was 0.34375 Hz. The same three-point smoothing kernel,

$$\mathbf{w} = \begin{bmatrix} 0.25 & 0.50 & 0.25 \end{bmatrix}, \tag{5}$$

was convolved with every auto-spectrum and cross-spectrum. Applying the same linear smoothing to all spectral quantities preserves their alignment and reduces isolated bin-to-bin variation. The smoothing setting and kernel were parameterized in the main analysis file.

For complex spectra, this paper uses the convention

$$S_{xy}(f) = \mathbb{E}[X^*(f)Y(f)], \tag{6}$$

where * denotes complex conjugation. This convention matters because the order of the signals determines the sign of phase.

2.5 Frequency bands

Band values were calculated after each subject's frequency-domain model was estimated. Each subject therefore contributed one value per band to each band-level statistical test; frequency bins were not treated as independent participants. Arithmetic metrics were averaged normally across bins. Phase was averaged circularly as described below. The lower band edge was included and the upper edge was excluded, except that the final band included its upper edge.

Table 1: Prespecified frequency bands used in the analysis.

Band	Abbreviation	Frequency range (Hz)
Very-very-low frequency	VVLF	$0.005 \leq f < 0.024$
Very-low frequency	VLF	$0.024 \leq f < 0.070$
Low frequency	LF	$0.070 \leq f < 0.200$
High frequency	HF	$0.200 \leq f \leq 0.350$

2.6 Single-input, single-output transfer functions

For a SISO model with input $x(t)$ and output $y(t)$, the frequency-domain model is

$$Y(f) = H_{xy}(f)X(f) + E(f), \tag{7}$$

where $H_{xy}(f)$ is the transfer function and $E(f)$ is unexplained output. Multiplying both sides by $X^*(f)$ and taking the expectation gives

$$\mathbb{E}[X^*Y] = H_{xy}\mathbb{E}[X^*X] + \mathbb{E}[X^*E], \tag{8}$$

$$S_{xy} = H_{xy}S_{xx}, \tag{9}$$

assuming the residual is uncorrelated with the input. The transfer function is therefore

$$H_{xy}(f) = \frac{S_{xy}(f)}{S_{xx}(f)}. \tag{10}$$

Separate SISO transfer functions were calculated for MAP-to-CBFV and $P_{\mathrm{ET}}CO_2$-to-CBFV:

$$H^{\mathrm{SISO}}_{\mathrm{MAP},y} = \frac{S_{\mathrm{MAP},y}}{S_{\mathrm{MAP,MAP}}}, \qquad H^{\mathrm{SISO}}_{CO_2,y} = \frac{S_{CO_2,y}}{S_{CO_2,CO_2}}. \tag{11}$$

Gain was the magnitude of the complex transfer function,

$$G_{xy}(f) = |H_{xy}(f)|\,. \tag{12}$$

Because CBFV was expressed as percent of its recording mean and both inputs were

expressed in mmHg, MAP and CO_2 gains are reported as percentage points of baseline CBFV per mmHg.

Wrapped phase was calculated explicitly as

$$\phi_{xy}(f) = \operatorname{atan2}(\operatorname{Im}\{H_{xy}(f)\}, \operatorname{Re}\{H_{xy}(f)\}), \tag{13}$$

where $\operatorname{Im}\{\cdot\}$ denotes the imaginary part. In standard notation, Equation 13 is the complex angle of H and returns a wrapped value in $[-\pi, \pi]$. Pairwise magnitude-squared coherence was

$$\gamma^2_{xy}(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)}. \tag{14}$$

The unexplained fraction was $1 - \gamma^2_{xy}$, and the corresponding residual power was

$$S_{ee}(f) = S_{yy}(f)\left[1 - \gamma^2_{xy}(f)\right]. \tag{15}$$

The CARNet ordinary-coherence reference for three Welch windows is 0.51 and for four windows is 0.41 [2]. These values were plotted only as visual benchmarks on ordinary SISO coherence figures. No frequency bin or participant was filtered on the basis of coherence. The ordinary-coherence reference was not applied to partial or multiple coherence because those statistics have different sampling behavior.

2.7 Two-input, single-output transfer functions

The MISO model treated MAP as $x_1(t)$, $P_{\mathrm{ET}}CO_2$ as $x_2(t)$, and normalized CBFV as $y(t)$:

$$Y(f) = H_1(f)X_1(f) + H_2(f)X_2(f) + E(f). \tag{16}$$

Multiplication by each conjugated input and expectation produces the spectral normal equations

$$S_{1y} = S_{11}H_1 + S_{12}H_2, \tag{17}$$

$$S_{2y} = S_{21}H_1 + S_{22}H_2. \tag{18}$$

In matrix form,

$$\underbrace{\begin{bmatrix} S_{11}(f) & S_{12}(f) \\ S_{21}(f) & S_{22}(f) \end{bmatrix}}_{\mathbf{S}_{xx}(f)} \underbrace{\begin{bmatrix} H_1(f) \\ H_2(f) \end{bmatrix}}_{\mathbf{H}(f)} = \underbrace{\begin{bmatrix} S_{1y}(f) \\ S_{2y}(f) \end{bmatrix}}_{\mathbf{S}_{xy}(f)}. \tag{19}$$

If the determinant

$$D(f) = S_{11}(f)S_{22}(f) - S_{12}(f)S_{21}(f) \tag{20}$$

is nonzero, the two transfer functions can be written as

$$H_1(f) = \frac{S_{22}S_{1y} - S_{12}S_{2y}}{D}, \tag{21}$$

$$H_2(f) = \frac{S_{11}S_{2y} - S_{21}S_{1y}}{D}. \tag{22}$$

The implementation did not calculate the inverse or Equations 21–22 directly. At each frequency, MATLAB solved

$$\mathbf{H}(f) = \mathbf{S}_{xx}(f) \backslash \mathbf{S}_{xy}(f), \tag{23}$$

using matrix left division. No ridge regularization or special singular-matrix fallback was applied in the present analysis. Gain and phase for H_1 and H_2 were calculated using Equations 12 and 13.

2.7.1 Multiple coherence

The fitted output power represented by both inputs is

$$S_{\widehat{yy}}(f) = \mathbf{S}_{xy}^H(f)\mathbf{H}(f), \tag{24}$$

where the superscript H denotes the conjugate transpose. Multiple coherence was calculated as

$$\gamma_{y\cdot 12}^2(f) = \frac{\mathrm{Re}\left\{\mathbf{S}_{xy}^H(f)\mathbf{H}(f)\right\}}{S_{yy}(f)}. \tag{25}$$

The MISO unexplained fraction was $1 - \gamma^2_{y\cdot 12}$ and residual power was $S_{yy}(1 - \gamma^2_{y\cdot 12})$. Because Equation 25 is calculated on the same data used to estimate the model, it was interpreted as in-sample linear representation, not out-of-sample predictive accuracy.

2.7.2 Partial coherence

Partial coherence measures the remaining linear association between one input and the output after removing the linear component associated with the other input. For MAP and CBFV conditional on $P_{\mathrm{ET}}CO_2$, the residual cross-spectrum and auto-spectra were

$$S_{1y\cdot 2} = S_{1y} - \frac{S_{12}S_{2y}}{S_{22}}, \tag{26}$$

$$S_{11\cdot 2} = S_{11} - \frac{S_{12}S_{21}}{S_{22}}, \tag{27}$$

$$S_{yy\cdot 2} = S_{yy} - \frac{S_{y2}S_{2y}}{S_{22}}. \tag{28}$$

The conditional MAP–CBFV partial coherence was then

$$\gamma^2_{1y|2}(f) = \frac{|S_{1y\cdot 2}(f)|^2}{\mathrm{Re}\{S_{11\cdot 2}(f)S_{yy\cdot 2}(f)\}}. \tag{29}$$

The conditional CO_2–CBFV partial coherence, $\gamma^2_{2y|1}$, was calculated by reversing the roles of inputs 1 and 2. Partial coherence was not described as the fraction of CBFV caused by a pathway. It is a conditional linear association after accounting for the other measured input.

2.7.3 Input coherence and conditioning

MAP–$P_{\mathrm{ET}}CO_2$ input coherence was

$$\gamma^2_{12}(f) = \frac{|S_{12}(f)|^2}{S_{11}(f)S_{22}(f)}. \tag{30}$$

The numerical condition number of the raw input spectral matrix was also stored:

$$\kappa(\mathbf{S}_{xx}(f)) = \|\mathbf{S}_{xx}(f)\|\,\|\mathbf{S}^{-1}_{xx}(f)\|. \tag{31}$$

A large value indicates that the transfer function solution can be sensitive to small spectral changes. However, the raw matrix contains MAP and $P_{\mathrm{ET}}CO_2$ in different units and power scales, so its condition number is also scale-dependent. It was used as a diagnostic, not an exclusion rule or a direct measure of physiological collinearity.

2.8 Phase representation and circular statistics

Wrapped phase was the primary representation for statistical analysis. Standard MATLAB unwrapping was used for the displayed unwrapped curves. The algorithm adds or subtracts 2π when it detects a discontinuity between neighboring frequency bins. This changes the displayed branch of the curve without changing its circular direction. An experimental custom unwrapping method remained available in the software but was not used for the reported results.

Linear averaging is inappropriate for wrapped angles because values just below π and just above $-\pi$ are close on a circle but far apart arithmetically. For phase values $\theta_1, \ldots, \theta_n$, the mean cosine and sine components were

$$C = \frac{1}{n}\sum_{j=1}^{n}\cos\theta_j, \qquad S = \frac{1}{n}\sum_{j=1}^{n}\sin\theta_j. \tag{32}$$

The circular mean direction was

$$\overline{\theta}_{\mathrm{circ}} = \mathrm{atan2}(S, C), \tag{33}$$

and the mean resultant length and circular standard deviation were

$$R = \sqrt{C^2 + S^2}, \qquad s_{\mathrm{circ}} = \sqrt{-2\ln R}. \tag{34}$$

These are standard directional-statistics definitions [12, 13]. Group phase means were first calculated circularly from the wrapped subject phases at each frequency. The circular mean curve was then unwrapped for display. The same circular standard deviation was retained around the wrapped and unwrapped group means because adding complete phase cycles does not change circular spread.

2.9 Statistical analysis

Statistical analysis was performed at two levels. Prespecified inference used one value per subject per band. Frequency-wise tests were exploratory and were performed separately at each frequency bin.

All inferential tests were two-sided. Confidence intervals were 95%, and the nominal significance level was 0.05 before the stated multiple-comparison adjustments.

2.9.1 Participant characteristics

Continuous participant characteristics were summarized as mean, standard deviation, and range and compared using Welch's unequal-variance two-sample t test [14]. Sex and medication counts were compared with Fisher's exact test. These descriptive comparisons were not treated as the primary TFA hypothesis family.

2.9.2 MISO versus SISO comparisons

MISO and SISO values came from the same subjects. Arithmetic outcomes were therefore compared using paired t tests on

$$d_i = x_i^{\mathrm{MISO}} - x_i^{\mathrm{SISO}}. \tag{35}$$

The null hypothesis was $H_0 : \mu_d = 0$. The standardized paired effect was

$$d_z = \frac{\overline{d}}{s_d}. \tag{36}$$

Because gain ratios were skewed in some bands, the median subject-level ratio

$$\widetilde{R} = \mathrm{median}_i \left(\frac{x_i^{\mathrm{MISO}}}{x_i^{\mathrm{SISO}}} \right) \tag{37}$$

and its interquartile range were reported as supporting effect descriptions. Paired phase differences were wrapped to the shortest angular difference and tested with a sign-randomization circular permutation test.

2.9.3 NC versus MCI comparisons

For arithmetic outcomes, NC and MCI were compared with Welch's two-sample t test. The test statistic was

$$t = \frac{\overline{x}_{\mathrm{NC}} - \overline{x}_{\mathrm{MCI}}}{\sqrt{s^2_{\mathrm{NC}}/n_{\mathrm{NC}} + s^2_{\mathrm{MCI}}/n_{\mathrm{MCI}}}}. \tag{38}$$

The null hypothesis was equality of group means at the tested band or frequency. Hedges' g described the standardized group difference with a small-sample correction [15]. Independent circular phase tests randomly reassigned pooled wrapped phase values to groups while preserving group sizes.

2.9.4 Permutation settings

Circular tests used 10,000 permutations. For each test, the raw permutation P value was

$$P = \frac{1 + \#\left(T_{\mathrm{perm}} \geq T_{\mathrm{observed}}\right)}{10{,}000 + 1}. \tag{39}$$

The added one prevents a zero P value. A fixed random seed of 2026 made the random permutations reproducible; it did not pull results toward zero or change the null hypothesis.

2.9.5 Multiple-comparison control

Raw P values answer one test at a time. Because many related bands and metrics were tested, the Benjamini–Hochberg procedure controlled the expected false discovery rate within prespecified analysis families [16]. For sorted raw values $P_{(1)} \leq \cdots \leq P_{(m)}$, the adjusted values were calculated as

$$P^{\mathrm{BH}}_{(i)} = \min_{j \geq i} \left\{ 1, \frac{m}{j} P_{(j)} \right\}. \tag{40}$$

Statistical significance was defined as $P^{\mathrm{BH}} < 0.05$. A value above 0.05 was interpreted as insufficient evidence to reject the null, not as proof that the groups or models were equivalent.

The primary MISO-versus-SISO family contained 16 gain comparisons: two groups, two pathways, and four bands. The primary NC–MCI family contained 16 MISO comparisons: two pathways, two metrics (gain and partial coherence), and four bands. The 32 secondary within-subject model comparisons (coherence and phase) were adjusted together.

The 36 secondary group comparisons (MISO phase, all SISO outcomes, and four pathway-balance comparisons) were adjusted together. The 24 input-coherence associations were one exploratory family. The nine physiological recording-characteristic tests exported by the workflow were also adjusted together, although Table 2 reports raw descriptive P values for participant and recording characteristics.

2.9.6 Exploratory frequency-wise tests

At each included frequency, NC and MCI were compared for MISO gain, partial coherence, and wrapped phase. MISO and SISO were also compared within each group for gain and phase. BH adjustment was applied across the bins belonging to each individual metric/pathway comparison curve. Thus, the adjusted P-value curve shows where one specific frequency curve provides evidence against its null hypothesis after accounting for the bins on that curve. The adjustment did not additionally control across every metric and pathway in the full project.

DRAFT NOTE: The current exploratory frequency-wise analysis includes the 0-Hz bin because the configured analysis range begins at 0 Hz, whereas the prespecified bands begin at 0.005 Hz. Mean removal should minimize the DC component, but the final paper should decide in advance whether to retain or exclude 0 Hz from the exploratory family and then rerun that analysis if the decision changes.

2.10 Software and reproducibility

The analysis was implemented in MATLAB R2025b using a single workflow for single-subject, batch, and synthetic-data runs. The same Welch spectra were passed to enabled SISO and MISO models, and the resulting subject-level values were passed to plotting, group summarization, statistics, and Excel export. The source code includes unit tests for preprocessing, spectral estimation, SISO and MISO calculations, circular statistics, BH adjustment, workflow settings, statistical analysis, and Excel export.

DRAFT NOTE: Add the final public repository URL, release/commit identifier, MATLAB toolbox requirements, data-availability statement, and instructions for obtaining any restricted participant data.

3 Results

3.1 Participants and recording characteristics

All 23 NC and 52 MCI recordings completed preprocessing and TFA. The NC and MCI groups did not differ in usable recording duration, number of Welch windows, mean MAP, mean $P_{\mathrm{ET}}CO_2$, mean CBFV, or CVRi (Table 2). Demographic matching was available for 23 NC and 51 MCI participants. Age, sex, body size, education, MMSE, blood pressure, heart rate, and medication counts were similar between groups. As expected from the clinical grouping, MoCA was lower in MCI. The available geriatric depression score was also higher in MCI, although values were low in both groups.

Table 2: Participant and recording characteristics.

Characteristic	NC	MCI	Raw P
Demographic and clinical characteristics			
Age, years	66.15 ± 7.14 [55.98–80.53]	64.78 ± 6.12 [55.11–78.75]	0.430
Female sex, n/N (%)	13/23 (56.5)	29/51 (56.9)	1.000
Height, cm	169.83 ± 7.98 [157.5–186.5]	168.35 ± 9.05 [143.5–187.8]	0.482
Weight, kg	78.59 ± 16.67 [50.6–110.6]	78.90 ± 14.71 [49.9–116.8]	0.938
Body mass index, kg/m^2	27.10 ± 4.65 [20.14–35.31]	27.74 ± 4.03 [20.27–35.81]	0.573
Education, years	16.22 ± 2.37 [12–20]	15.65 ± 2.39 [12–19]	0.345
MMSE, points	29.17 ± 0.94 [27–30]	29.18 ± 1.14 [24–30]	0.992
MoCA, points	27.35 ± 1.85 [24–30]	24.90 ± 2.72 [17–30]	< 0.001
Systolic BP, mmHg	129.65 ± 10.69 [113–153]	131.63 ± 11.29 [106–158]	0.474
Diastolic BP, mmHg	73.70 ± 8.90 [61–93]	74.61 ± 7.88 [56–91]	0.675
Heart rate, beats/min	72.35 ± 7.37 [59–86]	70.51 ± 10.04 [50–87]	0.381
Blood-pressure medication, n/N	8/23	16/47	1.000

Characteristic	NC	MCI	Raw P
Cholesterol medication, n/N	8/23	9/47	0.234
Geriatric depression score	0.57 ± 0.95 [0–3]	1.94 ± 1.78 [0–7][a]	< 0.001
Physiological recording and analysis characteristics			
Mean MAP, mmHg	90.73 ± 14.47 [68.64–116.97]	95.26 ± 13.90 [67.94–128.48]	0.214
Within-recording MAP SD, mmHg	4.11 ± 1.14 [2.31–7.19]	4.17 ± 1.44 [1.78–9.50]	0.843
Mean $P_{\mathrm{ET}}CO_2$, mmHg	36.68 ± 3.46 [29.79–42.50]	36.56 ± 2.85 [31.01–43.04]	0.890
Within-recording $P_{\mathrm{ET}}CO_2$ SD, mmHg	1.41 ± 0.68 [0.58–3.17]	1.32 ± 0.58 [0.48–3.31]	0.597
Mean CBFV, cm/s	46.43 ± 11.75 [31.98–81.99]	46.83 ± 10.17 [30.72–82.26]	0.888
Within-recording CBFV SD, cm/s	3.55 ± 1.16 [1.91–6.50]	3.35 ± 1.36 [1.35–8.66]	0.519
CVRi, mmHg/(cm/s)	2.03 ± 0.45 [1.06–2.89]	2.14 ± 0.60 [1.13–3.76]	0.413
Usable duration, s	297.35 ± 13.02 [270.5–347.8]	297.43 ± 14.19 [260.5–338.8]	0.980
Welch windows, n	3.04 ± 0.21 [3–4]	3.06 ± 0.24 [3–4]	0.795

Values are mean±SD [minimum–maximum] unless otherwise stated. Demographic values use 23 NC and 51 MCI participants because no demographic row matched MCI 302. Physiological recording values use all 23 NC and 52 MCI participants. [a]MCI $n = 50$ for the geriatric depression score. Medication denominators reflect available entries. BP, blood pressure; CBFV, cerebral blood flow velocity; CVRi, mean MAP divided by mean CBFV; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; SD, standard deviation.

3.2 Model dependence of gain

The primary methodological result was that simultaneous modeling changed pathway-specific gain, but the size and frequency pattern of that change were not identical for MAP and CO_2 (Table 3). In the MAP pathway, MISO gain was lower than SISO gain in VVLF for both

groups. NC also showed a small increase in MISO MAP gain in LF. The mean NC LF difference was 0.065, and the median subject-level MISO/SISO ratio was 1.028, indicating that the statistically detectable effect was small in absolute terms.

The more consistent changes occurred in the CO_2 pathway. NC MISO CO_2 gain was lower than SISO gain in LF and HF. MCI MISO CO_2 gain was lower in VLF, LF, and HF. The median MISO/SISO ratio in significant CO_2 comparisons ranged from 0.52 to 0.74. Thus, the separate CO_2-only model generally produced larger gain estimates than the conditional CO_2 estimate after MAP was included.

No MISO-versus-SISO phase comparison survived BH adjustment at the band level. Partial coherence and ordinary coherence were also compared, but these values were not treated as a test of model performance because they answer different questions.

FIGURE PLACEHOLDER: Conceptual difference between SISO and MISO transfer function analysis

Show MAP-to-CBFV and $P_{\mathrm{ET}}CO_2$-to-CBFV as two separate SISO models on the left. On the right, show MAP and $P_{\mathrm{ET}}CO_2$ entering one MISO model, with a dashed line between the inputs representing their cross-spectrum. State that SISO estimates pairwise associations, whereas MISO estimates each pathway conditional on the other measured input.

Figure 1: Conceptual difference between SISO and MISO transfer function analysis. Draft placeholder; replace this caption after the final figure is prepared.

Table 3: Primary within-subject comparison of MISO and SISO gain.

Group	Path	Band	MISO	SISO	MISO−SISO (95% CI)	Median ratio	BH P
NC	MAP	VVLF	1.092	1.332	−0.240 (−0.384, −0.096)	0.837	**0.0052**
NC	MAP	VLF	1.288	1.270	0.018 (−0.060, 0.096)	0.997	0.6370
NC	MAP	LF	2.111	2.046	0.065 (0.010, 0.120)	1.028	**0.0464**
NC	MAP	HF	2.343	2.331	0.012 (−0.036, 0.060)	1.014	0.6370
NC	CO_2	VVLF	4.127	4.370	−0.243 (−0.675, 0.189)	0.957	0.3149
NC	CO_2	VLF	2.684	3.170	−0.486 (−1.087, 0.116)	0.816	0.1444
NC	CO_2	LF	1.761	2.498	−0.737 (−1.067, −0.407)	0.662	**0.00052**
NC	CO_2	HF	2.695	4.944	−2.250 (−3.145, −1.354)	0.523	**0.00017**
MCI	MAP	VVLF	0.837	1.020	−0.183 (−0.290, −0.076)	0.844	**0.0038**
MCI	MAP	VLF	1.192	1.253	−0.061 (−0.116, −0.006)	0.971	0.0539
MCI	MAP	LF	1.779	1.769	0.010 (−0.028, 0.047)	0.994	0.6370
MCI	MAP	HF	2.034	1.995	0.039 (−0.001, 0.079)	1.015	0.0863
MCI	CO_2	VVLF	3.089	3.328	−0.239 (−0.493, 0.014)	0.982	0.0931
MCI	CO_2	VLF	2.066	2.850	−0.784 (−1.106, −0.462)	0.743	**0.000083**
MCI	CO_2	LF	1.599	2.892	−1.293 (−1.718, −0.869)	0.620	**0.0000021**
MCI	CO_2	HF	2.823	5.458	−2.635 (−4.275, −0.995)	0.682	**0.0052**

Values are group means. Gain units are percentage points of baseline CBFV per mmHg. Ratios were calculated within subject as MISO/SISO and summarized by the median. BH adjustment was performed across the 16 primary gain tests. Bold values are BH-adjusted $P < 0.05$. CI, confidence interval; HF, high frequency; LF, low frequency; MCI, mild cognitive impairment; NC, normal control; VLF, very-low frequency; VVLF, very-very-low frequency.

3.3 Coherence structure and pathway balance

Input coherence was modest across bands and similar between groups (Table 4). Conditional CO_2 partial coherence was comparable to or greater than MAP partial coherence in VVLF. MAP partial coherence became larger than CO_2 partial coherence from VLF through HF. This frequency pattern was present in both NC and MCI. Multiple coherence was between approximately 0.58 and 0.75 across bands, but it was interpreted as in-sample representation by both inputs rather than predictive validation.

The MAP-minus-CO_2 partial-coherence balance was negative in VVLF and positive in VLF, LF, and HF. The balance did not differ between NC and MCI in any band after BH adjustment. These findings indicate a similar broad frequency-dependent balance of conditional associations in both groups.

Table 4: Descriptive band coherence values.

Metric	Group	VVLF	VLF	LF	HF
Input MAP–CO_2 coherence	NC	0.314	0.250	0.216	0.209
	MCI	0.262	0.264	0.215	0.190
MISO MAP partial coherence	NC	0.335	0.560	0.592	0.656
	MCI	0.276	0.548	0.671	0.656
MISO CO_2 partial coherence	NC	0.409	0.346	0.221	0.225
	MCI	0.385	0.268	0.233	0.243
MISO multiple coherence	NC	0.645	0.711	0.681	0.733
	MCI	0.582	0.695	0.746	0.731
SISO MAP ordinary coherence	NC	0.388	0.542	0.584	0.647
	MCI	0.308	0.569	0.665	0.644
SISO CO_2 ordinary coherence	NC	0.456	0.300	0.198	0.209
	MCI	0.412	0.286	0.210	0.193
MAP minus CO_2 partial coherence	NC	−0.074	0.214	0.370	0.431
	MCI	−0.109	0.281	0.438	0.413

3.4 NC versus MCI comparisons

No prespecified band-level NC–MCI comparison of MISO gain or partial coherence survived BH correction (Table 5). The largest unadjusted gain difference was in the MAP LF band,

FIGURE PLACEHOLDER: Paired subject-level MISO and SISO gain comparisons

Use four panels for MAP VVLF, MAP LF, CO_2 LF, and CO_2 HF, or split MAP and CO_2 into two figures. Plot each participant's SISO and MISO band values connected by a line, with NC and MCI shown separately. Annotate the mean paired difference, 95% CI, median ratio, raw P, and BH-adjusted P. The figure should show effect size and variability rather than only a significance symbol.

Figure 2: Paired subject-level MISO and SISO gain comparisons. Draft placeholder; replace this caption after the final figure is prepared.

FIGURE PLACEHOLDER: Group coherence structure across frequency

For each group, show MAP–CO_2 input coherence, MAP partial coherence, CO_2 partial coherence, and multiple coherence as frequency curves with group mean and between-subject SD. In separate SISO panels, show ordinary MAP and CO_2 coherence with the CARNet reference line. Do not draw the ordinary-coherence threshold on partial or multiple coherence.

Figure 3: Group coherence structure across frequency. Draft placeholder; replace this caption after the final figure is prepared.

where NC had higher MISO gain than MCI (difference 0.332, 95% CI 0.024 to 0.640, Hedges' $g = 0.622$, raw $P = 0.0355$). After adjustment across the primary family, the BH-adjusted P value was 0.335. The largest partial-coherence difference was in the CO_2 VLF band (NC minus MCI 0.078, 95% CI 0.003 to 0.153, raw $P = 0.0418$, BH-adjusted $P = 0.335$). No band-level MISO phase comparison survived the secondary phase adjustment.

Table 5: Primary NC versus MCI comparisons of MISO gain and partial coherence.

Path	Metric	Band	NC	MCI	NC−MCI (95% CI)	Hedges g	BH P
MAP	Gain	VVLF	1.092	0.837	0.255 $(-0.140, 0.649)$	0.344	0.469
MAP	Gain	VLF	1.288	1.192	0.097 $(-0.092, 0.285)$	0.229	0.495
MAP	Gain	LF	2.111	1.779	0.332 $(0.024, 0.640)$	0.622	0.335
MAP	Gain	HF	2.343	2.034	0.309 $(-0.071, 0.689)$	0.477	0.344
MAP	Partial coh.	VVLF	0.335	0.276	0.059 $(-0.034, 0.153)$	0.330	0.469
MAP	Partial coh.	VLF	0.560	0.548	0.012 $(-0.089, 0.112)$	0.061	0.870
MAP	Partial coh.	LF	0.592	0.671	-0.079 $(-0.174, 0.015)$	-0.414	0.344
MAP	Partial coh.	HF	0.656	0.656	-0.0002 $(-0.086, 0.086)$	-0.001	0.996
CO_2	Gain	VVLF	4.127	3.089	1.038 $(-0.214, 2.291)$	0.493	0.344
CO_2	Gain	VLF	2.684	2.066	0.619 $(-0.426, 1.663)$	0.380	0.470
CO_2	Gain	LF	1.761	1.599	0.163 $(-0.291, 0.616)$	0.186	0.631
CO_2	Gain	HF	2.695	2.823	-0.128 $(-1.095, 0.839)$	-0.064	0.870
CO_2	Partial coh.	VVLF	0.409	0.385	0.024 $(-0.072, 0.121)$	0.112	0.759
CO_2	Partial coh.	VLF	0.346	0.268	0.078 $(0.003, 0.153)$	0.548	0.335
CO_2	Partial coh.	LF	0.221	0.233	-0.011 $(-0.042, 0.019)$	-0.184	0.631
CO_2	Partial coh.	HF	0.225	0.243	-0.017 $(-0.049, 0.014)$	-0.281	0.487

BH adjustment was performed across the 16 primary group tests. Gain units are percentage points of baseline CBFV per mmHg; coherence is unitless. No comparison met BH-adjusted $P < 0.05$. CI, confidence interval; coh., coherence; HF, high frequency; LF, low frequency; MCI, mild cognitive impairment; NC, normal control; VLF, very-low frequency; VVLF, very-very-low frequency.

3.5 Exploratory frequency-wise comparisons

The frequency-wise NC–MCI analysis identified one bin that survived BH adjustment within its comparison curve. At 0.1484375 Hz, MISO MAP gain was higher in NC than MCI (NC 2.462, MCI 1.710; difference 0.752, 95% CI 0.334 to 1.170; Hedges' $g = 0.959$; raw $P = 0.000813$; BH-adjusted $P = 0.0366$). Several neighboring bins had raw $P < 0.05$, but only the 0.1484-Hz bin survived the curve-wise correction. No frequency-wise group phase or partial-coherence bin survived its curve-wise BH correction.

Frequency-wise model comparisons showed broad MISO–SISO differences in CO_2 gain, especially in MCI, and more limited differences in MAP gain. These curves were consistent with the band results: simultaneous modeling had a stronger effect on CO_2 gain than on MAP gain. No frequency-wise phase comparison survived its curve-wise correction.

3.6 Input coherence and model-related gain changes

MAP–CO_2 input coherence was positively associated with the absolute MISO-minus-SISO gain change in several exploratory comparisons. In MCI, the associations included MAP VVLF ($\rho = 0.551$, BH-adjusted $P = 0.00038$), MAP LF ($\rho = 0.363$, BH-adjusted $P = 0.0338$), and CO_2 VVLF ($\rho = 0.365$, BH-adjusted $P = 0.0338$). In the pooled cohort, significant associations were present for MAP VVLF ($\rho = 0.537$, BH-adjusted $P = 2.46 \times 10^{-5}$), MAP VLF ($\rho = 0.293$, BH-adjusted $P = 0.0379$), MAP LF ($\rho = 0.283$, BH-adjusted $P = 0.0421$), CO_2 VVLF ($\rho = 0.375$, BH-adjusted $P = 0.0080$), and CO_2 VLF ($\rho = 0.347$, BH-adjusted $P = 0.0145$).

These results support the idea that the two models differ more when the inputs share more frequency content. However, they were not treated as independent biological validation because both input coherence and gain change are calculated from the same underlying spectral estimates.

3.7 Numerical conditioning

The raw input spectral matrix was more poorly conditioned at some high-frequency points. The median HF condition number was 10.9 in NC and 15.4 in MCI. Values greater than 100 occurred in 60 of 437 NC and 126 of 988 MCI HF frequency-by-subject observations. The maximum values were approximately 7,467 in NC and 17,366 in MCI. High values tended to

FIGURE PLACEHOLDER: Individual MISO values used in the NC–MCI comparisons

Plot individual NC and MCI values for the most relevant prespecified outcomes: MAP gain and partial coherence in VVLF/VLF/LF and CO_2 gain and partial coherence in VVLF/VLF/LF. Use a central estimate with a 95% CI, not mean ±SD error bars that extend beyond bounded coherence limits. Report raw and BH-adjusted P values. A supplementary figure can contain all four bands.

Figure 4: Individual MISO values used in the NC–MCI comparisons. Draft placeholder; replace this caption after the final figure is prepared.

FIGURE PLACEHOLDER: Exploratory frequency-wise NC–MCI comparison of MISO MAP gain

Top panel: NC and MCI group mean MISO MAP gain with between-subject SD or a clearly labeled 95% CI. Middle panel: NC-minus-MCI mean difference with a zero line. Bottom panel: raw and BH-adjusted P curves on a $-\log_{10}(P)$ axis so small values are visible, with a horizontal line at $-\log_{10}(0.05)$. Mark 0.1484 Hz and draw the prespecified band boundaries. State in the caption that BH correction was within this 45-bin curve and that the result is exploratory.

Figure 5: Exploratory frequency-wise NC–MCI comparison of MISO MAP gain. Draft placeholder; replace this caption after the final figure is prepared.

FIGURE PLACEHOLDER: Frequency-wise MISO versus SISO gain comparisons

Create separate NC and MCI figures or a compact multi-panel figure. For MAP and CO_2, show the MISO and SISO group curves, their paired difference, and the raw/BH-adjusted P curves. Use gain, not phase, in the main version because no phase comparison survived adjustment.

Figure 6: Frequency-wise MISO versus SISO gain comparisons. Draft placeholder; replace this caption after the final figure is prepared.

FIGURE PLACEHOLDER: Input coherence and the size of model-related gain changes

For all four bands, plot each subject's band-averaged MAP–CO_2 input coherence against the absolute MISO-minus-SISO gain change. Use separate MAP and CO_2 pathway panels, display NC and MCI with different colors, and annotate n, Spearman ρ, raw P, and BH-adjusted P. Keep the word "exploratory" in the title or caption.

Figure 7: Input coherence and the size of model-related gain changes. Draft placeholder; replace this caption after the final figure is prepared.

occur when $P_{\mathrm{ET}}CO_2$ power was low or MAP and $P_{\mathrm{ET}}CO_2$ powers were highly imbalanced.

Because the raw condition number depends on the units and scaling of both inputs, these observations do not establish severe physiological collinearity. They do show that some high-frequency MISO estimates may be sensitive to small spectral changes. No regularization or condition-number exclusion was applied.

DRAFT NOTE: Before submission, decide whether to add a scale-invariant conditioning diagnostic based on the normalized input spectral matrix and whether a prespecified ridge sensitivity analysis is needed. Do not add an arbitrary ridge constant after seeing the results. If regularization is used, justify its scale and show that the main conclusions are stable.

4 Discussion

4.1 Principal findings

This study asked two related but distinct questions. First, does simultaneous MAP and $P_{\mathrm{ET}}CO_2$ modeling materially change pathway-specific transfer function estimates? Second, after the pathways are modeled together, do the conditional estimates differ between NC and MCI? The answer to the first question was yes. The answer to the second was not at the prespecified band level.

The clearest methodological result was the reduction in MISO CO_2 gain relative to CO_2-only SISO gain. This reduction was present in NC at LF and HF and in MCI at VLF, LF, and HF. MISO MAP gain also differed from SISO MAP gain, but the changes were smaller and concentrated in VVLF. These findings show that the one-input and two-input models are not interchangeable. They do not show that the MISO model is universally superior. Rather, the models partition shared input–output information differently.

The main biological hypothesis was not supported. None of the prespecified band-level NC–MCI differences in MISO gain, partial coherence, phase, or pathway balance survived BH correction. This negative result is important. It indicates that accounting for spontaneous $P_{\mathrm{ET}}CO_2$ fluctuations did not reveal a broad conditional MAP-pathway deficit in the present MCI cohort. The finding is compatible with prior reports and systematic reviews showing mixed or absent group differences in spontaneous dCA metrics [5, 6, 7].

4.2 Why SISO and MISO gain differed

A SISO transfer function assigns all linear input–output cross-spectral information to one input. If MAP and $P_{\mathrm{ET}}CO_2$ are coupled, and both are associated with CBFV, the separate SISO estimates may include shared information. The MISO solution uses the full input spectral matrix and estimates both transfer functions at the same time. The conditional coefficient for one input can therefore decrease when variation shared with the other input is accounted for.

The strong reduction in CO_2 gain is consistent with this mechanism. The result does not mean that CO_2 is physiologically unimportant. Carbon dioxide remains a potent regulator of cerebral blood flow [8]. It means that the CO_2-only SISO gain was larger than the coefficient assigned specifically to $P_{\mathrm{ET}}CO_2$ after simultaneous MAP modeling. In other words, part of the pairwise $P_{\mathrm{ET}}CO_2$–CBFV relationship was shared with MAP in the two-input representation.

The pooled positive associations between input coherence and absolute gain change further support this interpretation. They are strongest in low frequency bands, where prior multivariable work has also shown that gas-related inputs can change pressure–flow estimates [9, 10]. Nevertheless, these associations should remain secondary because they are partly mathematically coupled to the same spectral matrix used to calculate the transfer functions.

4.3 Frequency-dependent pathway balance

The descriptive partial-coherence pattern was physiologically plausible but should not be overinterpreted. Conditional CO_2 coherence was comparable to or greater than conditional MAP coherence in VVLF. From VLF through HF, conditional MAP coherence was larger. This pattern suggests that slow CBFV variation contains an important CO_2-associated component, while MAP is more strongly associated with CBFV at higher frequencies in the analyzed range.

The pathway-balance result was similar in NC and MCI. It therefore provides a useful description of frequency-dependent association but not evidence of an MCI-specific shift. Moreover, partial coherence is not a direct measure of causal contribution. Unmeasured inputs, feedback, measurement noise, and nonlinear physiology can all affect the observed spectral relationships.

4.4 Interpretation of the NC–MCI results

The absence of BH-significant band differences should not be described as proof that NC and MCI are the same. The confidence intervals show the range of effects compatible with the current data. For example, NC MAP LF gain was higher than MCI before multiplicity adjustment and had a moderate Hedges' g. The adjusted analysis, however, did not provide enough evidence to declare that result across the prespecified family. The correct interpretation is that the study did not establish a robust band-level group difference.

The isolated 0.1484-Hz MAP-gain difference deserves follow-up but not a primary claim. It survived BH adjustment across the bins of its own curve and was surrounded by several smaller raw P values, which is more credible than a completely isolated numerical spike. Still, the frequency-wise analyses were exploratory, multiple metric/pathway curves were examined, and the finding was not a prespecified single-frequency hypothesis. It should therefore be used to guide a future targeted analysis or replication rather than to redefine the main conclusion of this study.

4.5 Coherence as a benchmark rather than a filter

The present analysis retained all frequency bins and used ordinary SISO coherence as a visual benchmark. This choice separates data description from data deletion. With three Welch windows, the CARNet ordinary-coherence reference was 0.51 [2]. Much of the CO_2 ordinary coherence, especially outside VVLF, was below this reference. Consequently, the high- frequency CO_2 gain and phase estimates should be interpreted cautiously even when MISO and SISO gains differed statistically.

The ordinary threshold does not transfer directly to partial or multiple coherence. Applying the same number to those metrics would imply a sampling distribution that has not been established here. Showing the ordinary line only on SISO coherence plots is therefore more defensible than filtering the full analysis or drawing the same line on every coherence statistic.

4.6 Phase findings

No phase comparison survived multiplicity correction. This is not surprising given the circular variability of phase, the limited number of Welch windows, and abrupt branch changes

near $-\pi$ and π. Circular averaging prevents the specific error of treating those boundaries as far apart, and testing the wrapped values avoids making inferential results depend on the chosen display branch. Unwrapped curves remain useful for visualization, but they should be shown as the unwrapped circular mean with the same circular standard deviation.

4.7 Multiple coherence and prediction

Multiple coherence was generally higher than either single pairwise coherence because two inputs were fitted to the same output on the same data. This is expected and does not by itself demonstrate that MISO predicts CBFV better in new data. A meaningful prediction comparison would estimate the spectra or model on a training segment and evaluate reconstructed CBFV on a separate test segment. Most recordings in this cohort provided only three 128-second Welch windows, leaving little independent data for both stable estimation and testing.

Cross-validation is therefore better suited to a future supplemental analysis using longer recordings. A 20-second holdout after the 256-second minimum would contain relatively little low-frequency information and would not provide a strong test of VVLF or VLF prediction. Longer recordings or an experimental design with repeated baseline epochs would allow a more defensible training/test separation.

4.8 Numerical stability

The MISO solution was obtained without regularization. This keeps the current analysis transparent, but some raw spectral matrices were poorly conditioned, especially at high frequencies with low $P_{\mathrm{ET}}CO_2$ power. The largest gain changes also occurred in higher-frequency CO_2 bands, so numerical stability and physiological interpretation cannot be separated completely.

A future sensitivity analysis should first normalize the input spectral matrix so that conditioning is not dominated by the different units of MAP and $P_{\mathrm{ET}}CO_2$. If regularization is then needed, the ridge scale should be defined before viewing group results and justified relative to the normalized matrix or measurement-noise level. The smallest ridge that produces stable solutions across a prespecified sensitivity range would be preferable to an arbitrary constant. The current paper should not claim that such a sensitivity analysis was performed unless it is actually added.

4.9 Strengths

This study has several strengths. First, it directly compares SISO and MISO estimates within the same participants, so model-related changes are not confounded by different cohorts. Second, MAP, $P_{\mathrm{ET}}CO_2$, and CBFV share one spectral estimation step, and identical smoothing is applied to all auto- and cross-spectra. Third, phase means and dispersion use circular rather than linear statistics. Fourth, inferential tests use one band value per subject and do not treat frequency bins as independent subjects. Fifth, raw and BH-adjusted P values, confidence intervals, standardized effects, and median gain ratios are exported with the subject-level values used in each test. Finally, the workflow is parameterized from one main file and includes tests for its major numerical and export components.

4.10 Limitations

Several limitations affect interpretation. The study was observational, so the transfer functions describe association and do not establish causal MAP- or CO_2-mediated effects. $P_{\mathrm{ET}}CO_2$ is a surrogate for arterial carbon dioxide and may be influenced by ventilation, dead space, and measurement delay. Transcranial Doppler measures velocity rather than volumetric flow and assumes that the insonated arterial diameter is sufficiently stable [17]. Only baseline spontaneous fluctuations were analyzed, which may provide limited input power compared with repeated squat–stand or controlled CO_2 challenges.

Most recordings produced only three Welch windows. This meets the configured minimum but limits spectral precision, raises the ordinary-coherence reference, and prevents strong cross-validation. The $P_{\mathrm{ET}}CO_2$ startup rule was designed for the available exports and should be checked in new datasets. The MISO model included two inputs, but cerebral blood flow is affected by other variables, including oxygen, cardiac output, autonomic activity, metabolism, and neurovascular coupling. Feedback between MAP, ventilation, and the cerebral circulation also challenges a simple feed-forward linear model.

The raw input-matrix condition number was scale-dependent, and no ridge sensitivity analysis was performed. Frequency-domain smoothing makes neighboring bins more similar, so frequency-wise P values should not be read as independent discoveries. Finally, the NC and MCI sample sizes were unequal, one MCI participant lacked demographic data, and the diagnostic composition of MCI must be described more fully from the parent protocol.

4.11 Future work

The next study should prioritize longer recordings or repeated baseline epochs, prospectively defined model-stability criteria, and out-of-sample prediction. Controlled MAP oscillations and controlled CO_2 challenges could provide stronger and more separable inputs than spontaneous baseline variation. A normalized conditioning analysis and prespecified regularization sensitivity test would clarify the stability of high-frequency coefficients. The 0.1484-Hz MAP-gain finding should be tested as a targeted hypothesis in an independent cohort rather than searched for again across the same frequency range.

Future biological analyses could also test associations with continuous cognitive scores, MCI subtype, vascular risk, medication exposure, or imaging markers rather than relying only on a binary NC–MCI comparison. Those analyses should be defined before testing and should account for age, sex, education, and other relevant covariates when the sample size supports them.

5 Conclusion

Simultaneously modeling MAP and $P_{\mathrm{ET}}CO_2$ materially changed transfer function gain estimates, with the largest and most consistent changes in the CO_2 pathway. This supports the methodological argument that separate MAP-only and CO_2-only SISO models can combine information that is partitioned differently in a two-input model. In contrast, the current data did not show a robust prespecified band-level difference in conditional MAP- or CO_2-associated CBFV dynamics between NC and MCI. The results therefore support pathway separation as a useful analysis framework, while they do not support a broad MCI-specific impairment under resting spontaneous conditions. The exploratory difference at 0.1484 Hz and the high-frequency conditioning findings provide clear directions for replication and sensitivity analysis.

Declarations

Ethics approval and consent to participate

DRAFT NOTE: Insert the institutional review board name and approval number, and state how written informed consent was obtained.

Data availability

DRAFT NOTE: State whether de-identified physiological data can be shared, whether access requires a data-use agreement, and where the synthetic example data and analysis code will be deposited.

Code availability

The MATLAB analysis workflow will be made available at [repository URL and archived release to be added].

Funding

[Funding information to be added.]

Conflicts of interest

[Conflict-of-interest statement to be added.]

Author contributions

[Author contribution statement to be added.]

Acknowledgments

[Acknowledgments to be added.]

A Recommended figure set and placement

The following set keeps the main paper focused while preserving the statistical evidence:

1. **Main Figure 1:** Conceptual SISO-versus-MISO schematic (Figure 1).
2. **Main Figure 2:** Paired subject-level MISO-versus-SISO gain comparisons (Figure 2).
3. **Main Figure 3:** Coherence structure across frequency (Figure 3).

4. **Main Figure 4:** Individual NC and MCI MISO band values (Figure 4).

5. **Main or supplemental Figure 5:** Exploratory 0.1484-Hz MAP-gain result with adjusted P curve (Figure 5).

6. **Supplemental Figure S1:** Full frequency-wise MISO-versus-SISO gain results (Figure 6).

7. **Supplemental Figure S2:** Input coherence associations (Figure 7).

8. **Supplemental Figure S3:** Conditioning diagnostics (Figure 8).

Representative single-subject raw signals and subject-specific transfer function curves could be added as a supplemental quality-control figure. They are useful for showing what the input data and model output look like, but they should not replace group distributions or confidence intervals.

B Items that must be resolved before submission

1. Confirm participant diagnosis, recruitment, exclusions, IRB, consent, acquisition devices, posture, and recording protocol from the parent study.

2. Decide whether the 0-Hz bin remains in exploratory frequency-wise tests.

3. Decide with the PI whether normalized conditioning and a prespecified ridge sensitivity analysis are needed. The main reported solve is currently unregularized.

4. Decide whether the primary family should remain all four bands or emphasize VVLF/VLF based on the biological hypothesis. This must be decided prospectively for any rerun and reported transparently.

5. Confirm whether the Geriatric Depression Scale belongs in the main participant table and whether medication variables are sufficiently complete for publication.

6. Confirm how to describe MCI 302, which has TFA results but no matching row in the demographic workbook.

7. Add a permanent code release and data-availability statement.

8. Select the target journal and reformat title page, abstract length, headings, tables, references, and figure limits.

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FIGURE PLACEHOLDER: Input spectral-matrix conditioning diagnostics

Suggested supplementary figure: condition number versus frequency for each group, plus condition number versus $P_{\mathrm{ET}}CO_2$ power and $\mathrm{MAP}/P_{\mathrm{ET}}CO_2$ power ratio. Clearly state that the plotted raw condition number is scale-dependent.

Figure 8: Input spectral-matrix conditioning diagnostics. Draft placeholder; replace this caption after the final figure is prepared.